

Question Number	Answer	Mark
3(a)(i)	The only correct answer is B behavioural <i>A is not correct because migration is an example of a behavioural adaptation</i> <i>C is not correct because migration is an example of a behavioural adaptation</i> <i>D is not correct because migration is an example of a behavioural adaptation</i>	(1)

Question Number	Answer	Additional guidance	Mark
3(a)(ii)	1328	accept 1.328×10^3 ignore 1.3×10^3 and 1.33×10^3	(1)

Question Number	Answer	Additional guidance	Mark
3(b)	An explanation that makes reference to the following points: - population size would have reduced (1) (as fewer milkweed plants mean) fewer eggs will {be laid / hatch / survive} / fewer offspring produced / fewer butterflies will be available to reproduce (in next generation) / fewer {places / plants} to lay eggs (1) (fewer reproducing butterflies) could result in reduced genetic diversity (of butterfly population) (1)	ignore no eggs will be laid / no place to lay eggs ignore fewer eggs are fertilised unless linked to future generation(s) e.g. fewer eggs fertilised means fewer butterflies to reproduce (next year) accept decreased reproduction rate ignore mutation could result in increase in genetic variation ignore inbreeding might occur ignore references to genetic diversity of milkweed	(3)

Question Number	Answer	Additional guidance	Mark
3(c)(i)	{many genes / genes at different loci} contribute to {the rate of egg laying / this characteristic} (1)	ignore alleles unless correctly qualified accept {many genes / genes at different loci} contribute to the characteristic / trait / phenotype	(1)

Question Number	Answer	Additional guidance	Mark
3(c)(ii)	- graph showing normal distribution (1) - both axes labelled correctly (1)	accept peak shifted to left or right accept suitable labels e.g. x axis = phenotype / characteristic / rate of egg laying y axis = frequency / number of {hens / individuals} ignore population <u>Example of graph</u>  accept graphs which do not start at 0	(2)